Information Management and Systems Engineering

Topic Exchange for Bachelor Theses

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1 CONCEPTUAL MODELING

1.1 Business Model Outline & Description

This project aims at implementing an online exchange for bachelor thesis topics. The overall intention is therefore to provide a central platform, where lecturers can offer open topics, which can then be easily browsed and selected by students based on their interests. Taking into account the preferences specified by the students, the system can then definitively pair up topics with students after the deadline for the choices passed.

Therefore, users, encompassing instructors and students, can first register with their university email-address and their full name, then set their username and password. Additionally, students also have to provide their matriculation number and study program code, while instructors optionally also give their contact information. Some of the instructors are administrators, which means that they have elevated privileges, most importantly they can instruct the system to start the matching process. All users are identified by an unique UserID[1], which is automatically generated by the system.

Instructors are also affiliated with exactly one research group. Research groups are identified by their GroupID, but described by their unique name and their research profile[2].

After successful registration, an instructor may post multiple thesis topics (identified by TopicIDs), which would be supervised by them. In doing so, they have to provide a unique title and description of it. Furthermore, they can refer to literature references, which contain at least a title and optionally also an author, a year of publication and a link to the referenced resource. References hence only exist within one particular topic and are identified within by a reference number, which start at one for each topic and are incremented for every additional reference.

It is also possible to indicate that a thesis topic belongs to one or more categories, which for example represent research disciplines or used technologies. Categories are identified by a CategoryID and have a name, as well as a short description that contains a concise explanation of the category. One example could be the category "C++" with the description "Bachelor projects in this category will be implemted with the C++ programming language.". On top of that, categories may form hierarchies, that is, a category can be a subcategory of one other category.

Students can then search through the thesis topics, filtering them by their category, the instructor or the instructor's research group, and bookmark the ones they find enticing. After having decided on topics, which they might want to work, they can choose several. Finally, students are required to allocate a limited number of points among their choices. Higher point allocations signify greater priority and will factor into the system's matching process when pairing students with topics. In addition, the system saves a timestamp for each choice, in order to allow tie-breaking strategies based on "first-come, first-serve". In the end, an administrator can start the matching process, in the course of which the system will assign topics to students based on different rules, the rule which led to an assignment will be given as its reason.

1 All attributes of the form `<entity name>ID` that will be mentioned in the remainder of this outline are also automatically generated unique identifiers for the respective entities.
2 The research profile is a short description of the research foci and activities of the group.

1.2 Entity-Relationship Model

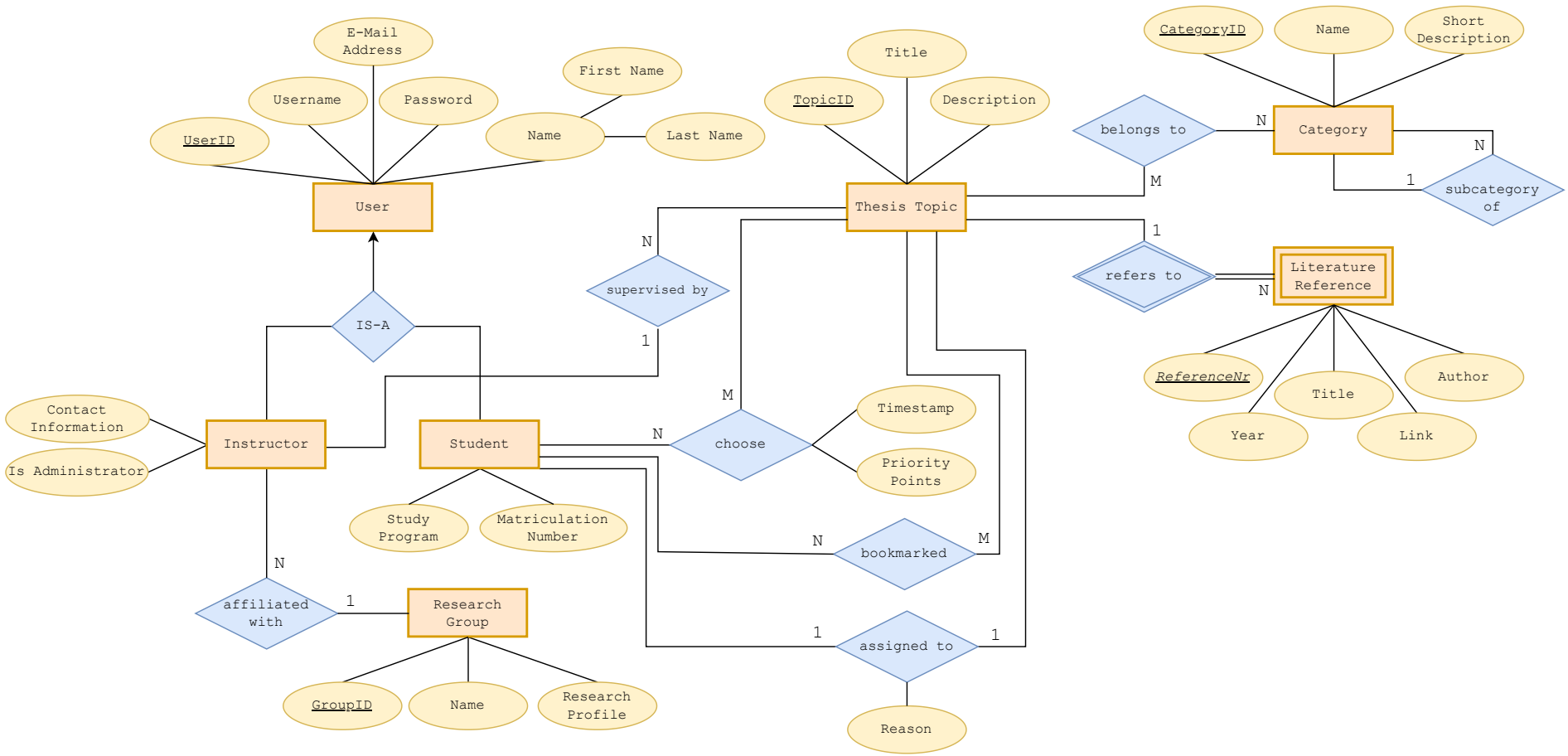

Figure 1: Entity-Relationship Diagram

2 USE-CASE DESIGN

2.1 Creation of a Thesis Topic

OBJECTIVE

An instructor posts a new thesis topic.

SHORT DESCRIPTION

An instructor, who wants to supervise some topic, can submit it to the topic exchange, providing its title, description and categories, as well as literature references.

PRECONDITIONS

The instructor is logged in.

EXPECTED EXECUTION

The instructor clicks the button for the addition of a new topic. Then the instructor enters the topic's title and the topic's description on the page that opens up. Then they search through the available categories and select fitting ones. Finally, they add literature references, by providing the title and optionally the other details.

POSTCONDITIONS IN CASE OF SUCCESS

The new thesis topic was created successfully, stored in the database, and is therefore now visible to students browsing the available thesis topics.

POSTCONDITIONS IN CASE OF ERRORS

Duplicate Title: If a topic with the given title already exists, nothing will be stored to the database and an error message is displayed. The instructor has to retry the addition with a new title.

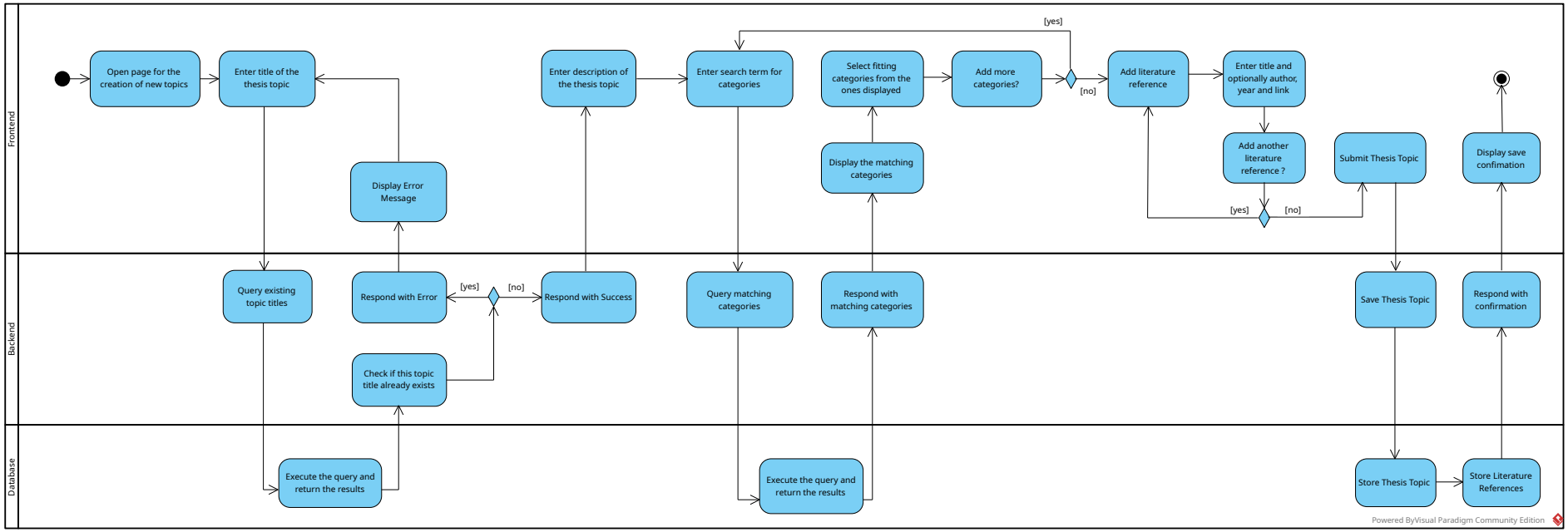

Figure 2: UML Activity Diagram: Creation of a Thesis Topic

2.2 Selection of a Thesis Topic

OBJECTIVE

A student applies for one of the available thesis topics.

SHORT DESCRIPTION

A student selects a thesis topic in which he would be interested and distributes his preference points according to his intention.

PRECONDITIONS

The student is already logged in and has identified the wanted thesis topic in the search results list.

EXPECTED EXECUTION

1. The student selects his desired thesis topic from the search results list by clicking on the box.
2. Next he selects the "Apply for thesis topic" button in the pop-up info screen.
3. A new window pops up asking the user to select a number of preference points from those that are still available and to confirm his choice.
4. After his confirmation, the user's preference list of applied thesis topics is displayed in his profile.

POSTCONDITIONS IN CASE OF SUCCESS

The thesis topic application has been saved, it is visible to the student in his profile and can be taken into account in the final assignment of thesis topics.

POSTCONDITIONS IN CASE OF ERRORS

- If the student has already applied for this thesis topic, the details view is displayed again after the error message is displayed.
- If the student has entered an invalid number of preference points and has no more points available for the application for another topic, the details view is displayed again after the error message is displayed.
- If the student has entered an invalid number of preference points, but there are still points available to distribute, he will be asked to enter the preference points again after the error message is displayed.

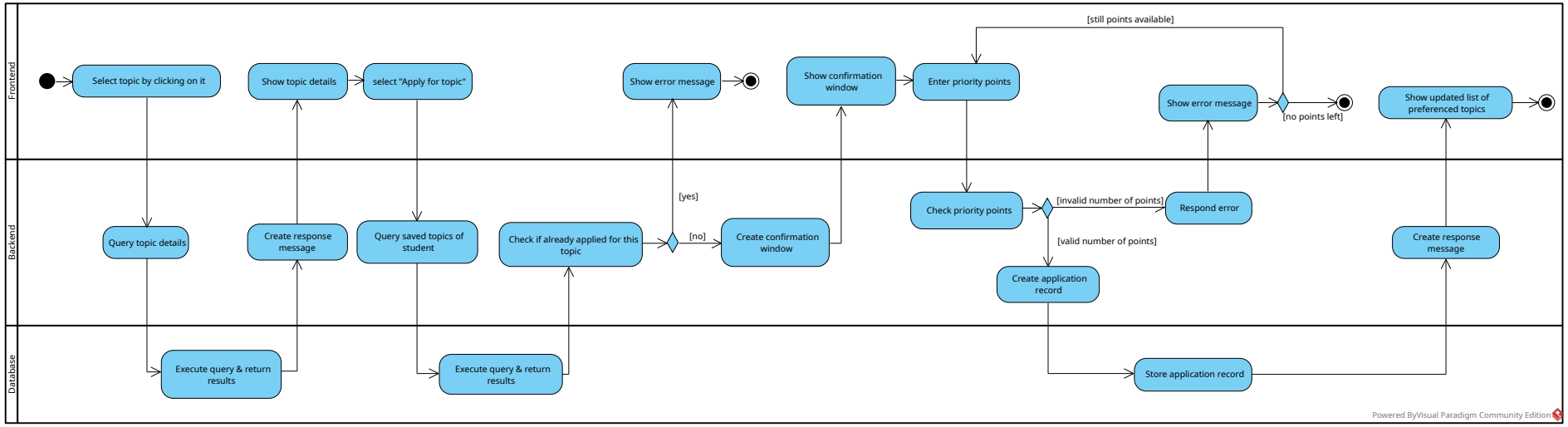

Figure 3: UML Activity Diagram: Selection of a Thesis Topic

2.3 Setting a bookmark

OBJECTIVE

A student sets a bookmark to save a thesis topic.

SHORT DESCRIPTION

A student who is interested in several thesis topics first bookmarks them so that he can decide later which one he would like to apply for.

PRECONDITIONS

The student is already logged in and has already identified thesis topics which seem interesting to him in the search results but wants to think about which one(s) he wants to apply for later.

EXPECTED EXECUTION

The student clicks on the thesis topic he wants to bookmark for later decision. The thesis topic information screen pops up. Now the student clicks the "Bookmark" button to save it.

POSTCONDITIONS IN CASE OF SUCCESS

The thesis topic has been saved and can be viewed in the student's profile in the bookmark section and can also be removed again if desired.

POSTCONDITIONS IN CASE OF ERRORS

If the thesis topic has already been bookmarked, the user is asked whether he wants to remove the bookmark or cancel the process.

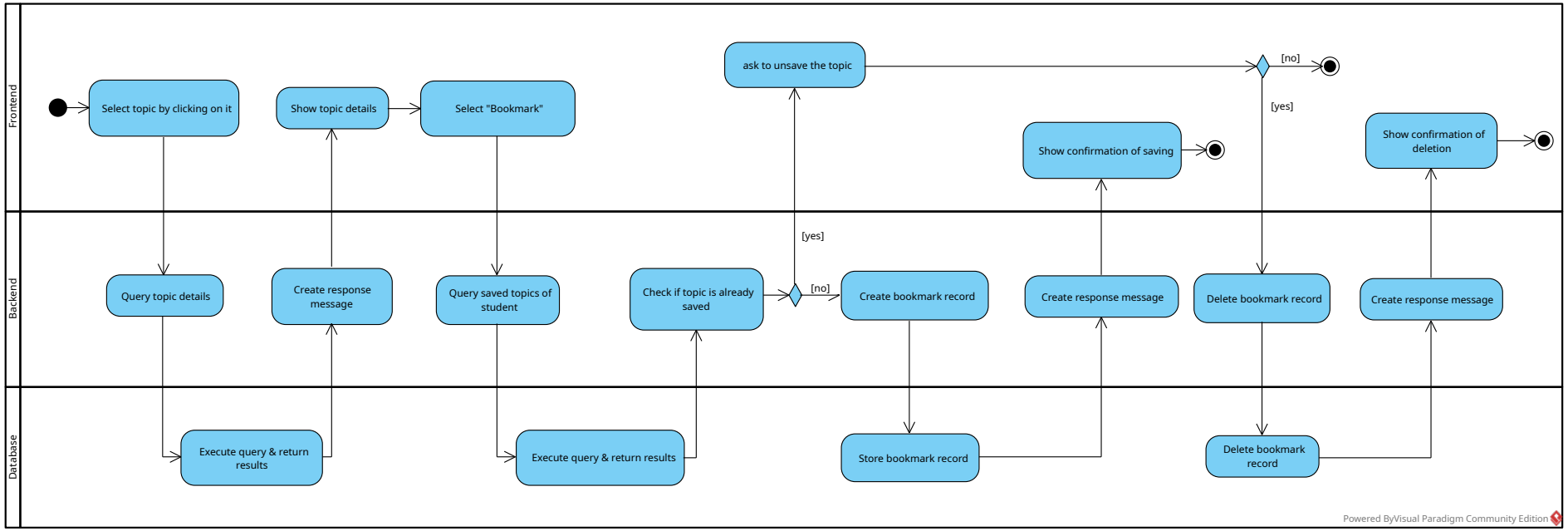

Figure 4: UML Activity Diagram: Setting a bookmark

2.4 Assignment of Thesis Topics

OBJECTIVE

The system matches thesis topics with students.

SHORT DESCRIPTION

An instructor with administrator rights performs the process of assigning thesis topics to the students with the help of the system.

PRECONDITIONS

An instructor is logged in and has administrator rights.

EXPECTED EXECUTION

The administrator kicks off the matching process, whereupon the system automatically assigns topics based on the students' preferences, trying to resolve conflicts between the choices in the best interest of the students. The administrator is presented with the assignment of topics, as well as the reasons for the decisions, and can adapt it. Afterwards, the systems stores the assignments in the database.

POSTCONDITIONS IN CASE OF SUCCESS

The assignment of topics is completed and stored in the database along with the reasons.

POSTCONDITIONS IN CASE OF ERRORS

Not Enough Topics: If there are fewer topics than students, the assignment is not performed and the database not changed. Instead, the respective error message is displayed, and the administrator has to find additional supervisors before retrying the assignment process.

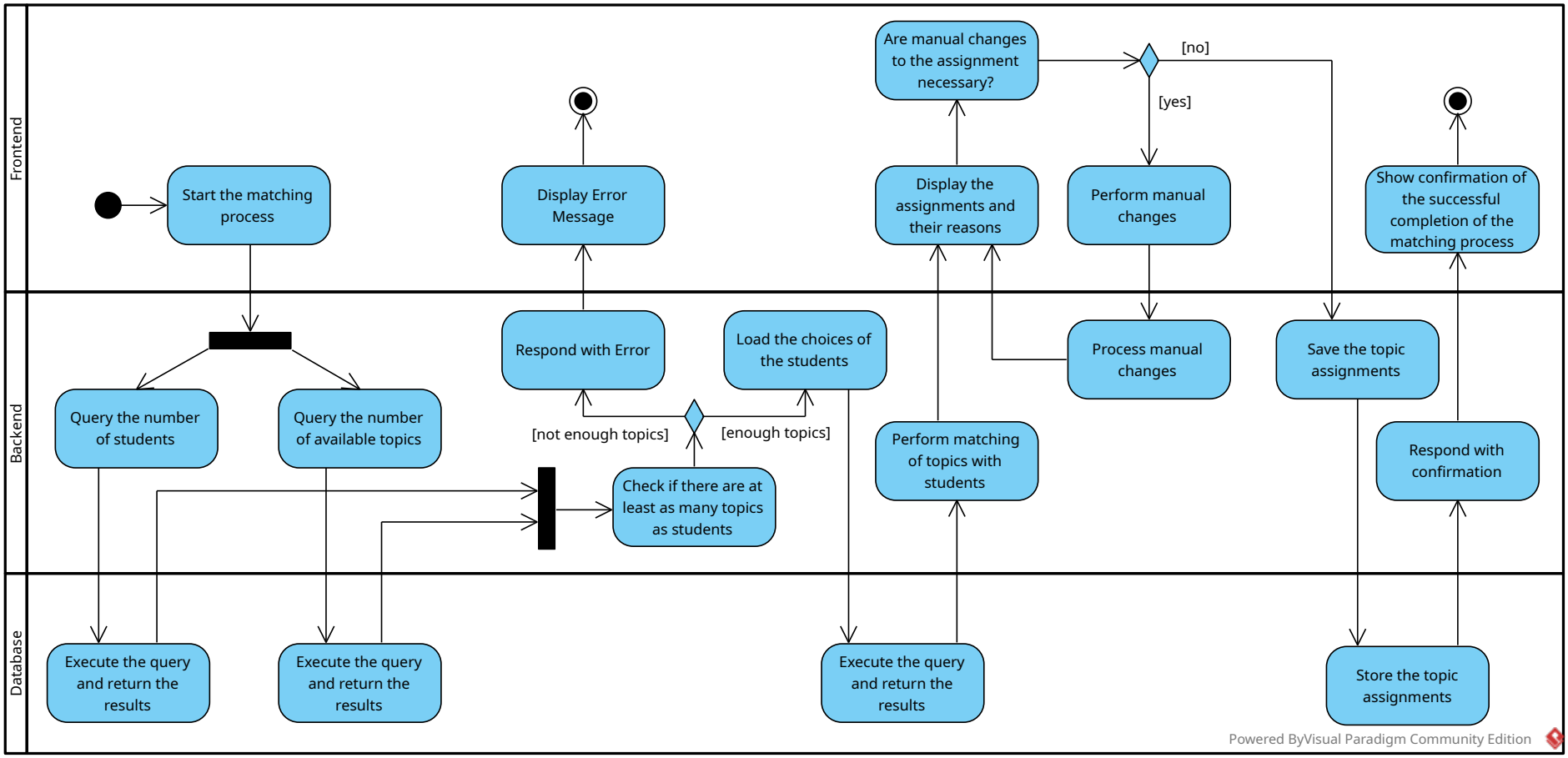

Figure 5: UML Activity Diagram: Assignment of Thesis Topics